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Current Economic Affairs Year XXVII, No. 93 – September / December 2017

ISSN 2250-754X online

Causal relationship between the producer price index and the consumer price index: Ecuador Case

Causality relation between the producer price index and the consumer price index. Ecuador Case

Victor Quinde-Rosales* and Rina Bucaram-Leverone**

* Faculty of Agricultural Economics of the Agrarian University of Ecuador (Ecuador)
** Faculty of Agricultural Economics of the Agrarian University of Ecuador (Ecuador)
˙ vquinde @uagraria.edu.ec
¨ rbucaram@uagraria.edu.ec

Summary

This research document, employing inductive reasoning, evaluated the causal relationship between the Producer Price Index (PPI) and the Consumer Price Index (CPI) over the period from January 1998 to December 2016. Using an empirical-analytical paradigm, the Augmented Dickey-Fuller Unit Root (AFR) test was used, a Vector Autoregressive (VAR) model was generated, and the Granger causality test was performed. The results demonstrate a positive trend and seasonality in the data for the variables. A two-variable VAR model with an optimal number of lags of fourteen – VAR2(14) – was obtained, and the causality test was performed, demonstrating a bidirectional relationship between the two indices.

Keywords: price index; causality; unit root; vector autoregression.

JEL Code: C32; C40; E31; E50.

Abstract

The present document is an investigation with a type of inductive reasoning. It evaluated the relationship of causality between the producer price index (IPP), and the consumer price index (IPC) in a period from January 1998 to December 2016. The unit root test Dickey-Fuller Augmented (DFA) was used under an empirical- analytic paradigm, an autoregressive vector-VAR model was generated and the Granger causality test was performed. The results show a positive trend and seasonality in the data of the va-riables, a VAR model of two variables was obtained with a number of optimal remnants of fourteen VAR2 (14) to which the causality test was performed, demonstrating a bi - directionality of both indices.

Key words: price index; causality; unit root; autore-gressive vector.

JEL Code: C32; C40; E31; E50.

1. Introduction

Price fluctuations are one of the main triggers of market instability. According to Kalmanovitz (2003), cited by Pérez et al. (2014), these fluctuations distort accurate information, generating inefficient allocation of resources, variations in competitiveness, and changes in the freedom of economies.

In Ecuador, as in most countries, the producer price index (PPI) and the consumer price index are determined.

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(CPI), understanding that the first is the monthly evolution of producer prices of goods offered to the domestic market at its first stage of commercialization, while the second measures changes over time at the general level of prices, corresponding to the final consumption of goods and services of households of the different income levels, residing in the urban area of the country.

The importance of the aforementioned indicators lies in calculating inflation and using them as inputs for updating tariffs and prices, adjusting wages, and indexing a large amount of economic information. Although Ecuador does not have its own currency for monetary policy, it is the State's responsibility, through public policy, to maintain low and predictable inflation that contributes to savings, investment, economic growth, and job creation.

It is under this premise that the present study intends to establish whether the PPI allows anticipating changes in the trend of the CPI activity throughout the economic cycle, understanding the causality of the first index in the dynamics of the second.
If this hypothesis is verified, the Ecuadorian State can establish price regulation policies for the producer that generate an effect on the price for the consumer and on inflation.

The design of this research seeks to characterize the causal relationship that exists between the studied indices by reviewing the literature of authors who have developed similar studies, and then presenting the methodological framework to be used in the development of the study: Subsequently, the results of the research arising from the implementation of an autoregressive vector model are presented, which show the simultaneity of the aforementioned price indices, and finally, the Granger test is performed to establish the existing causality.

The conclusions section contains an analysis of the context in which the research results were developed. The references section lists the works cited in this document that were used to outline the study and its findings.

2. Development

2.1 Literature Review

Comparing time variables is important for economic analysis.
For decision-making, short-term indicators become important by demonstrating, through the application of studies, the constant changes over time, whether historical or current, generating forecasts of their behavior in the future (INEC, 2016).

The Producer Price Index (PPI) has been calculated by the National Institute of Statistics and Censuses (INEC) since 1998, although its calculation began in 1997 at the request and development of the Central Bank of Ecuador (BCE) with the advice of the International Monetary Fund (IMF), using 1995 as the base year.

This index is considered by the INEC (National Institute of Statistics and Censuses) to be an early indicator of inflation, as it measures the evolution of prices of goods produced for the domestic market. Therefore, we can argue that the PPI (Producer Price Index) reflects the price behavior of the first link in the marketing chain for the Agriculture, Livestock, Forestry, Fishing, and Manufacturing sectors. These prices are obtained through interviews and surveys of representatives of manufacturing companies, agricultural producers in wholesale markets, and agricultural production units that make up the sample segments of the Continuous Agricultural Area and Production Survey (ESPAC); and with fishing vessels at landing ports and coves (INEC, 2016).

We can define the producer price as the money received by the producer from the buyer for each unit of a good produced, excluding value-added tax (VAT) or any other indirect tax. Additionally, it excludes transportation costs that would be billed separately.

The current methodological framework of the IPP uses 2005 as its base year; this indicator is part of the Production Indicators System (SIPRO), which evaluates production.

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national in three dimensions: real production value, prices of products generated for the domestic market and labor supply from the perspective of companies. This indicator bases its coverage on 44 of the 71 branches of economic activity and 136 of 278 products existing in the System of National Accounts of Ecuador, whose classification is correlated with the International Standard Industrial Classification (ISIC) and Central Product Classification (CPC) (INEC, 2016).

The Consumer Price Index (CPI) is a national index calculated monthly. This indicator aims to assess changes in the final consumption prices of goods and services for households residing in urban areas of the country, taking into account nine cities represented by a fixed consumption structure representative of urban household spending, as generated through the National Survey of Income and Expenditure of Urban and Rural Households (ENIGHUR).

This indicator is the most important measure of overall inflation, as it reflects final household consumption expenditures, largely representing the final spending of the Ecuadorian economy. We can say that the CPI allows us to monitor changes in the general price level of the economy.

Under this criterion, the objective of this indicator within the economy is defined as an index that serves as a measure for adjusting the monetary values of certain payments or stocks, such as wages, housing rents, social benefits, interest, taxes, and the value of monetary assets and liabilities; thus making the CPI an important indicator in the design of economic policy. Similarly, this indicator is related to the national accounts in establishing the balance between supply and consumption in the country's economy. It is under this premise that the CPI becomes an important macroeconomic indicator with economic, social, legal, and political influence (INEC, 2015).

The consumer price index is constructed using information on the consumption expenditure of the national urban population, without excluding any segment or stratum of the population.
The cities considered for the decision-making process The areas covered are Guayaquil, Esmeraldas, Machala, Manta, Santo Domingo de los Colorados, Cuenca, Loja, Ambato, Quito, and the urban population of their respective provinces, achieving a population coverage of 83 percent under the ENIGHUR sampling frame (INEC, 2015).

Both Martínez et al. (2012); as well as Gómez together with Rodríguez (2012) and with Navarro (2012) agree in the determination of a broad study carried out to establish the causality of the PPI and the CPI, understanding this as empirical evidence that does not determine the direction of causality between both indices in each of the countries studied by the authors.

Colclough et al. (1982) use the Sim (1972) and Granger (1969) tests for causality between producer and consumer prices for the case of the United States, these indicate that causality goes from the consumer to the producer.
Jones (1986) evaluates the same case in the United States, showing as evidence a bidirectional causality in prices.

Caporale et al. (2002) analyzed the relationship between the PPI and the CPI of the G7 countries over a study period between January 1976 and April 1999. For the development of this study, a bivariate VAR model was used with an analysis system of causality in the sense of Granger (1969). The results show a single direction of PPI to CPI in France and Germany. In the case of Italy, Japan, the United Kingdom and the United States, it occurs in both directions; while in Canada no causality was found.

Ghazali et al. (2008) used monthly data from Malaysia, within a study period from January 1986 to April 2007, seeking to establish the link between the PPI and the CPI. These authors used two types of tests: the error correction model of Engle and Granger (1987) and the causality test of Toda and Yamamoto (1995). Both tests found a causal relationship between the PPI and the CPI.

Liping et al. (2008) established the relationship between the PPI and the CPI using monthly data from China, finding a long-term relationship between both indices and a bidirectional causality, being stronger

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that which goes from producer prices to consumer prices. To establish the relationship of the indicators, the approach of Johansen (1988, 1991) was used, while the test of Toda and Yamamato (1995) was used to determine the type of causality between the baskets.

Sidaoui et al. (2010) applied an error correction model and Granger's causality test (1969), finding that in the case of Mexico, the monthly variations of the indices under study exhibit a long-term cointegration dynamic, and the causality is from the PPI to the CPI. Aviral et al. (2010), in their study, sought to determine the causality between the PPI and the CPI in India, finding evidence of bidirectionality in the short and long term.

Akcay (2011) examined the causality between the PPI and the CPI of five European countries, using seasonally adjusted monthly data from August 1995 to December 2007. The causality test used was that of Toda and Yamamoto (1995), which found unidirectional causality from the PPI to the CPI in the cases of Finland and France, and bidirectional causality between the indices in Germany. No significant causality was detected for the Netherlands and Sweden.

Muhammad et al. (2012) used causality in the Granger (1969) sense, under the frequency domain proposal of Lemmens et al. (2008), between the consumer price index and the wholesale price index (WPI) of Pakistan using the study period from 1961 to 2010; finding that the causal relationship between these indicators

It varies according to the frequency domain. Additionally, they discovered that the CPI Granger-causes the PPI at all frequency levels, reflecting long, medium, and short-term cycles in which the aforementioned causality persists.

In two of his publications, Gómez seeks to establish the causal relationship between the PPI and the CPI by incorporating structural changes in the data for Mexico, along with Navarro (2012), and by studying the member countries of the North American Free Trade Agreement (NAFTA), along with Rodríguez (2012). In both cases,

The study aimed to assess the seasonality of the data using the Dickey-Fuller (1979 and 1981), Phillips-Perron (1988), and Lee-Strazicich (2003) tests for Mexico. Simultaneously, it sought to establish causality by applying the Granger test (1969) to both cases and the Toda and Yamamoto (1995) test for Mexico. The study periods varied accordingly. For Mexico, the period was from January 1994 to February 2012, while for the NAFTA study, the evaluation period was from January 1957 to April 2011. The results for Mexico reflect seasonal data with a unidirectional causality from the CPI to the PPI; whereas the NAFTA study suggests a bidirectional causality between the aforementioned indices.

Huertas et al. (2000) sought to establish a relationship between the PPI and the CPI of Colombia within a study period from December 1990 to December 1999 using Granger's causal relationship (1969) by altering the indicators in two instances. In the first instance, they excluded non-common components, services in the CPI and intermediate consumption in the PPI, resulting in a bidirectional causality; while in the second instance, they generated more disaggregated levels by separating the aforementioned indicators into food and medicine, finding links between the indicators.

Finally, Martínez et al. (2012) propose to explore the relationship between the PPI and the Colombian CPI in a study period from January 1991 to June 2012. To obtain their results, the authors proposed to establish which of the two indicators is the leading index under the Banerji test (1999), determining that the PPI can anticipate the evolution of the CPI for one or even several months.

We can argue that the series of authors who have evaluated the relationship between the producer price index and the consumer price index use as methodology in most of them the vector autoregressive (VAR) model or its version of the Bayesian vector autoregressive (BVAR) model together with the study of causality in the Granger sense or the Toda and Yamamoto modification.

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2.2 Methodology

This research is framed within a type of inductive reasoning with the application of econometric tests to measure the probability of the arguments established in relation to the causality of the PPI and the CPI. The research development process is outlined as a single-method process with an empirical-analytical paradigm, following Bacon (1949) cited by Duque (2015), which allows for the most faithful and neutral reflection possible of the reality of the research conducted.

For the research design of this document, a monthly time series database was used, with an evaluation period from January 1998 to December 2016, for the spliced variables1 producer price index (PPI)2 and consumer price index (CPI)5, whose database was obtained from the National Institute of Statistics and Censuses (INEC). The methodological proposal of the research proposes an analysis that determines the stationarity of the variables through the use of the Augmented Dickey-Fuller test (ADT) to avoid obtaining spurious results4 in regressions with non-stationary time series.

The Dickey-Fuller (DF) test simulates the critical values of a selection of different sample sizes, understanding that the stochastic process underlying the data is an autoregressive AR(1)5 model (Novales, 1993).
The DFA test considers other autocorrelation schemes, avoiding non-compliance with the white noise condition6 for the residuals.

1. Statistical procedure to make series with different base years comparable.
2. Historical series spliced to link the base 1995=100 to the base: 2015 = 100; excluding the Mining sector.
3. Historical series spliced to link the base: 2004=100 to the base: 2014=100.
4. Spurious is the statistical relationship of two or more variables that still do not have a causal or coincidental relationship.
5. One-lag autoregressive model.
6. Random signal that presents values at two different times that are not statistically correlated.

Stock et al. (2012) mention that the DFA contrast for an autoregressive unit root contrasts the null hypothesis H0: ÿ = 0 against the alternative hypothesis H1: ÿ < 0 in the following regression:

$$\Delta y_t = \beta_1 + \beta_2 t + \delta y_{t-1} + \sum_{i=1}^{m} \alpha i \Delta y_{t-i} + \varepsilon_t$$

Understanding that:
H0: The variable is non-stationary; it has a unit root
H1: The variable is stationary; it has no unit root

The DFA does not have a standard distribution, so the usual critical values of the normal distribution cannot be used for its unit root test. Instead, a specific set of critical values must be used based on the distribution of the DFA statistic under the null hypothesis (Stock et al., 2012). Wooldridge (2010) explains that because the alternative hypothesis of stationarity implies that ÿ < 0 in the equation described above, the test is one-sided. Therefore, the value of the t-statistic must be less than (more negative than) the one-sided critical values to reject H0.

or null hypothesis.

Additionally, MacKinnon (1996) estimated a set of critical values which allow calculating the critical values of the DFA test for different sample sizes and numbers of variables, understanding that this value must be less than 5% (0.05) to reject the null hypothesis.

To establish a relationship between the PPI and the CPI, the Vector Autoregressive (VAR) model is used. Sims (1980), cited by Gujarati et al. (2010), argues for the existence of simultaneity among the set of variables to be treated under equal conditions, without distinction between exogenous and endogenous variables. It is called a vector autoregressive model when there is a lagged value of the dependent variable and a vector of two or more variables.

Novales (2011) describes that the application of VAR presents its usefulness when there is a coincidence between a group of variables, these relationships must be generated over a certain number of periods (Pérez et al., 2014).

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For the application of the VAR model, the following is taken Taking into consideration the following equations:

$$M_{1t} = \alpha + \sum_{j=1}^{k} \beta_j M_{t-j} + \sum_{j=1}^{k} \gamma_j R_{t-j} + \mu_{1t}$$

$$R_t = \alpha' + \sum_{j=1}^{k} \theta_j M_{t-j} + \sum_{j=1}^{k} \gamma_j R_{t-j} + \mu_{2t}$$

The following hypotheses are proposed:

Hypothesis 1. The relationship between the variables is less than one year, being considered short-term.

Hypothesis 2. The relationship between the variables is greater than one year, being considered long-term.

The existence of a relationship between variables does not, in itself, constitute a synonym for causality, nor does it specify its type of orientation. Furthermore, as Gujarati (2010) points out, a philosophical conflict persists, arguing that the future cannot predict the past. Under this criterion, the Granger test interprets causality as follows: if a variable X causes, in Granger terms, variable Y, then changes in X should predict changes in Y, and vice versa.

In terms of regression, we can argue that including past values or lags of X in the regression of Y would contribute to a more meaningful prediction of Y. Based on this premise, we can state that X, in Granger terms, causes Y. Stock et al. (2012) concur, stating that X is a useful predictor of Y.

Wooldridge (2010) adds that Granger causality is a limited notion that describes how past values of X are useful in predicting future values of Y after being controlled for past values of Y.

According to the Granger test, the following hypotheses are proposed.

Hypothesis 1. Unidirectional causality from IPP to IPC; rejection of hypothesis 1.

Hypothesis 2. Unidirectional causality from IPC to IPP; rejection of hypothesis 2.

Hypothesis 3. Bidirectional causality between the variables; both hypotheses are rejected.

Hypothesis 4. Independent variables; both hypotheses are not rejected.

To test the hypotheses, the Fischer probability (Prob) is taken into account. If the p-value is within the minimum rejection threshold, less than 5% or 0.05, one of the hypotheses is rejected.
Based on this criterion we can describe the possible results (Table 1).

Table 1. Hypothesis Testing Granger Test

Hypothesis	Hypothesis 1	Hypothesis 2	Hypothesis 3	Null hypothesis 4
IPC no cause to IPP	< 5%	> 5%	< 5%	> 5%
IPP no cause to IPC	> 5%	< 5%	< 5%	> 5%

For the analysis of the data pursued in this document, the econometric software Eviews 9 was used, which presented the results of the contrasts for the development of the results.

3. Results and Discussion

The Producer Price Index (PPI) and the Consumer Price Index (CPI) show a positive trend during the study period from January 1998 to December 2016.

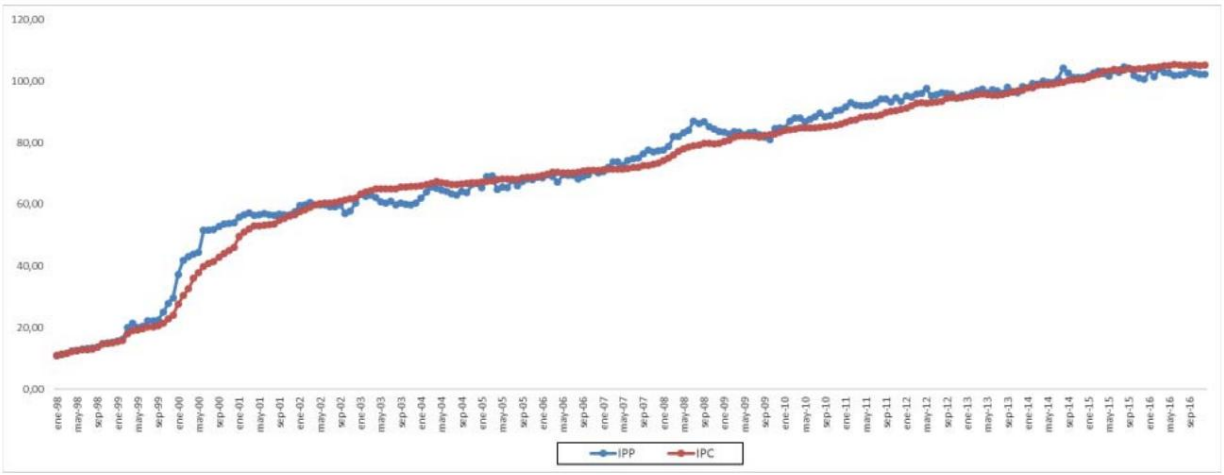
The aforementioned behavior of the variables is similar except for periods such as October 1999 to September 2001, March 2003 to February 2004, June 2007 to March 2009, and February 2010 to August 2012, where the difference in the PPI is not noticeable, but persists during the study period. Huertas et al. (2000) state that this possible difference is due to the different measurement of prices, understanding that the CPI is a purchase price while the PPI is a supply price (Figure 1).

To determine whether the trend was deterministic or stochastic, the Augmented Dickey-Fuller (ADF) test was performed. The results of this test show that the null hypothesis was rejected for both indicators, so we can conclude that both the PPI and the CPI are series that do not have a unit root and are therefore stationary (Table 2).

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Fig. 1. Producer Price Index and Consumer Price Index



Source: own elaboration

Table 2. Unit Root Test of the PPI and CPI

Variable	DFA Test		Prob*
	Test critical values: 5% level	t-Statistic	
IPP	-2.874	-3.293	0.016
IPC	-2.874	-3.013	0.035

*MacKinnon (1996) one-sided p-values

For the vector autoregressive (VAR) estimation process, it is necessary to establish the maximum lag length over which the relationship between the IPP and IPC variables is determined understanding this criterion as the optimal lag.
To do this we will use the likelihood ratio test, the Final Prediction Error criteria and the Akaike, Schwarz and Hannan-Quinn statistics. These criteria show a discrepancy when establishing the lag.

The last two criteria highlight the second lag as the optimal one, while the first three tests highlight the fourteenth lag (Table 3).

Table 3. Selection Criteria for the Order of Delays – VAR IPP IPC

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-1.4567	THAT	4231.7	14.026	14.058	14.039
1	-4.5598	1972.58	0.2912	4.4422	4.5384	4.4811
2	-4.3055	49.6510	0.2370	4.2360	4.3965*	4.3009*
3	-4.2687	7.10622	0.2377	4.2391	4.4638	4.3300
4	-4.243.	4.75427	0.2412	4.2537	4.5425	4.3705
5	-4.1960	9.07032	0.2394	4.2461	4.5991	4.3889
6	-4.1765	3.66162	0.2442	4.2658	4.6830	4.4345
7	-4.1411	6.55580	0.2453	4.2703	4.7517	4.4650
8	-4.0899	9.40921	0.2428	4.2595	4.8051	4.4801
9	-4.0771	2.17644	0.2495	4.2865	4.8962	4.5330
10	-4.0294	8.72237	0.2475	4.2783	4.9522	4.5508
11	-4.0072	3.97246	0.2519	4.2953	5.0334	4.5937
12	-3.9702	6.48001	0.2528	4.2983	5.1006	4.6227
13	-3.8230	25.6326	0.2281	4.1952	5.0616	4.5455
14	-3.7297	16.0488*	0.2169*	4.1440*	5.0746	4.5203
15	-3.7065	3.95541	0.2206	4.1601	5.1549	4.5623
16	-3.6916	2.51141	0.2262	4.1842	5.2432	4.6124
17	-3.6661	4.23669	0.2296	4.1982	5.3214	4.6523
18	-3.6397	4.33154	0.2329	4.2113	5.3987	4.6914
19	-3.6341	0.91873	0.2411	4.2443	5.4959	4.7504
20	-3.6182	2.55547	0.2471	4.2675	5.5832	4.7995

* indicates lag order selected by the criterion. LR: sequential modified LR test statistic (each test at 5% level). FPE: Final predic-tion error. AIC: Akaike information criterion. SC: Schwarz information criterion. HQ: Hannan-Quinn information criterion.

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Table 4. Parameter Estimation of the Equations – VAR IPP IPC

Parameter	R2	F-statistic	Akaike AIC	Black SC
IPP (-2)	0.997	19095.3	3.436	3.512
IPC (-2)	0.999	224387.9	1.008	1.083
IPP (-14)	0.996	1.869.6	3.521	3.977
IPC (-14)	0.999	25691.4	0.968	1.425

To establish the optimal lag, we must evaluate the parameter estimates of the equations generated for the PPI and CPI at 2 and 14 lags. The aim is to verify the pre-accuracy of the adjusted VAR forecast. The analysis of the coefficients collectively to establish significance using the F-test registers a high value, indicating statistically significant lag terms (Table 4).

A similar case occurs with the coefficient of determination or R2 which both in the two model Lags such as the one with fourteen lags show a high coverage value when predicting future results (Table 4). The Akaike criteria and the Schwarz criteria cannot be evaluated in light of the previous assessment of the optimal lag (Table 3).

Given the ambiguity of the results in the VAR estimation process between the variables studied, a correlogram analysis was performed. as a residual test, which evaluates the behavior of the residuals within the confidence intervals; understanding the ability to evaluate the residuals in terms of relevance or explanatory power of the given model. The correlogram highlights as the appropriate model the one that presents 14 lags in each variable (Figure 2).

Understanding that the optimal lag of the VAR between the PPI and the CPI is fourteen lags – VAR2(14) it is determined that the indicators present a response to shocks between variables and among themselves propagating in the long term.

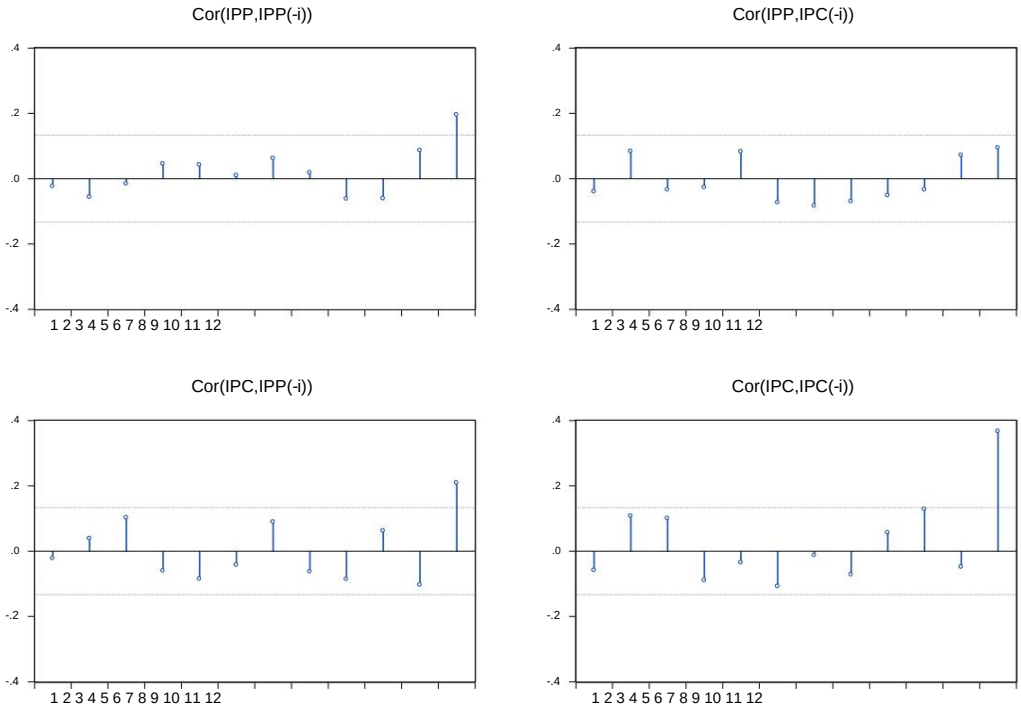
To establish causality, a The Granger test, whose results show bidirectional causality, rejects the null hypothesis that the IPC does not Granger-cause the IPP and that the IPP does not Granger-cause the IPC. This is evidenced by observing the p-values in Table 5, that is, taking into account the Fisher probability (Prob), which is at the minimum rejection value.

Table 5. Granger Causality Test – VAR IPP IPC of 14 Lags

Null hypothesis	Granger Test		
	Chi-sq	df	Prob.
IPC does not cause IPP	3.297	14	0.0029
IPP does not cause IPC	2.509	14	0.0336

Figure 2: Correlograms

Autocorrelations with 2 Std.Err. Bounds

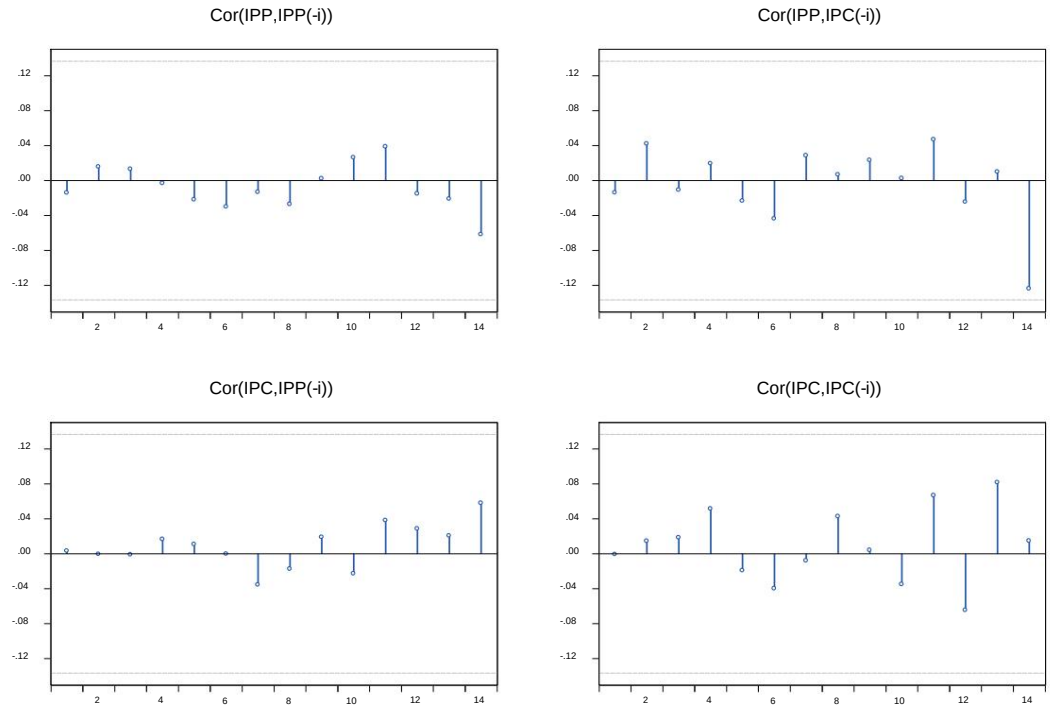


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Figure 2: Correlograms (continued)

Autocorrelations with 2 Std.Err. Bounds



4. Conclusions

Establishing the causal relationship between the producer price index (PPI) and the consumer price index (CPI) is of interest to the development of Ecuador's public policy, which aims to maintain low inflation for the benefit of family savings, investment, and job creation.

There are non-significant differences between the PPI and the CPI which may be due to the methodology of price measurement; as well as the generalization of the study which does not differentiate the behavior of the food market, the health market, or the services market.

In studying the causal relationship between the PPI and the CPI, it was determined that both indicators are positive trend, stationary, and deterministic; that their monthly basis generated a VAR2(14) model, which explains the relationship between the variables studied and between themselves as an optimal lag. In terms of causality, it was determined that, according to the Granger test, the model presents bidirectionality in terms of Granger causality.

The results continue to contribute to the empirical development of the topic, given the diversity of responses found in other studies. Further research on this topic, using diverse methodologies, is needed to strengthen the empirical development of the subject.

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